

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Project Name	Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau	Maximum Marks	4 Marks
Team ID	XXXXXX	Date	7th July 2026

1. Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

1.1 System Process Flow

- Data Ingestion:** The system extracts raw consumer feedback, social media trends, and product reviews from e-commerce platforms and brand APIs.
 - Data Preprocessing & Sentiment Analysis:** Raw textual and numerical data are cleaned, filtered, and structured using a Python NLP pipeline to generate sentiment scores and trend metrics.
 - Data Warehouse Storage:** Structured insights and processed trend data are loaded into a secure SQL database store.
 - Tableau Integration:** Tableau connects directly to the structured data store to fetch live and historic metrics.
 - Dashboard Visualization:** End users (Brand Analysts and Administrators) interact with dynamic, visual dashboards to analyze cosmetics trends and consumer preferences.

DATA FLOW MAPPING (LEVEL 0 - INDUSTRY STANDARD)

Source / Entity	Data Flow	Process	Target / Store
E-commerce & Social Media	Raw Review & Trend Data	1. Collect & Extract Data	Data Store (Raw Cache)
Data Store (Raw Cache)	Unstructured Text	2. Clean & Perform Sentiment Analysis	Data Store (Processed Insights)
Brand Analyst	Query Filters & Requests		Tableau Dashboards

Source / Entity	Data Flow	Process	Target / Store
		3. Generate Tableau Visualizations	
Data Store (Processed Insights)	Structured Metrics	3. Generate Tableau Visualizations	Tableau Dashboards
Administrator	System Configurations / User Roles	4. Manage Users & Access Control	Data Store (User Registry)

2. User Stories

The following table outlines the comprehensive user stories, epics, acceptance criteria, priorities, and release plans for the *Cosmetics Insights* platform across all standard user personas.

User Type	Functional Req (Epic)	ID	User Story / Task	Acceptance Criteria	Priority	Release
Brand Analyst (Web User)	Dashboard Navigation	USN-1	As a brand analyst, I can view the main trend dashboard so that I can see top-performing cosmetics categories at a glance.	Dashboard loads within 3 seconds and displays the top 5 trending categories.	High	Sprint-1
	Filtering & Search	USN-2	As a brand analyst, I can filter consumer insights by brand, region, and demographic data to customize my market reports.	All charts dynamically update automatically when a filter is applied.	High	Sprint-1
	Sentiment Analysis	USN-3	As a brand analyst, I want to see sentiment breakdown charts (Positive, Neutral, Negative) for specific ingredients to mitigate product risks.	Pie charts and time-series sentiment lines are correctly rendered in Tableau.	Medium	Sprint-2
	Exporting Reports	USN-4	As a brand analyst, I can export Tableau visualizations as PDF or PowerPoint reports to share with executives.	Export button generates high-resolution files containing the current view.	Low	Sprint-2
Customer Care Executive	Feedback Tracking	USN-5	As a customer care executive, I can view real-time negative spikes in consumer sentiment regarding product formulas.	Alerts or highlights show up on the dashboard when negative sentiment exceeds 20%.	High	Sprint-1
	Issue Resolution	USN-6	As a customer care executive, I can drill down into specific low-rated reviews from the dashboard to understand product complaints.	Clicking a chart segment opens the corresponding anonymized feedback records.	Medium	Sprint-2
Administrator	User Access Control	USN-7	As an administrator, I can manage user permissions and access levels to ensure data confidentiality.	Admin portal allows adding/removing users and modifying roles.	High	Sprint-1
	Data Source Config	USN-8	As an administrator, I can configure and refresh the web-scraping schedules and database connections.	Data refresh triggers successfully manually or	High	Sprint-1

User Type	Functional Req (Epic)	ID	User Story / Task	Acceptance Criteria	Priority	Release
				automatically via cron schedule.		
	Audit Logging	USN-9	As an administrator, I want to view system activity logs to monitor who accessed specific proprietary dashboards.	Logs document user ID, timestamp, and actions taken accurately.	Low	Sprint-2